

The universal rung profile $\Delta(\ell)$: a lemma candidate

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Abstract

The quasi-null ladders of the fifteen truncated Weil forms share a spacing profile $\Delta(\ell)$ to $\pm 12\%$. The dilated-Slepian model for that profile has no well-defined exponent and is not the lemma. What the thirty-three rungs actually say is sharper and weaker: after a short rise, spacings *saturate* at $\Delta_\infty \approx 16$ nats, independently of the L -function. The $\sqrt{\ell}$ fit is a two-parameter compromise with a ramp-then-plateau, not an asymptotic. We state the universality statement as a lemma candidate in pure harmonic analysis (approximation numbers of windowed Fourier restriction, after scaling out λ_0), record the factorization into one arithmetic speed $s(\chi)$ times one window-invariant profile, and isolate the two assertions a proof must supply: (i) scale invariance of the deflated problem past the infrared turning point; (ii) character-blindness of the leftover operator. No claim is made that the lemma is proved.

1 The object

Fix $\mu = e^L$ and a real primitive character χ (or ζ). Let $Q_{\chi,\mu}$ be the semi-local Weil form on even test functions supported in $I_L = [-L/2, L/2]$, assembled as in [?]. Write $0 < \lambda_0 \leq \lambda_1 \leq \dots$ for its eigenvalues and

$$\ell_k = -\ln \lambda_k, \quad \Delta_k = \ell_{k+1} - \ell_k = \ln \frac{\lambda_k}{\lambda_{k+1}}.$$

Call ℓ_k the *depth* of rung k and Δ_k its *spacing* (nats). The ground-state law $\ell_0 = s(\chi)\mu + b(\chi)$ is *not* part of this note: $s(\chi)$ is arithmetic. The question here is the internal architecture of one spectrum, at fixed window.

Across 33 rungs of fifteen L -functions (characters at $\mu = 38$, ζ at $\mu = 16$ in a depth-adequate basis) the spacings collapse by level bands [?, ?]:

level band ℓ	Δ	sample
[0, 15)	10.9 ± 1.5	ζ interleaves with χ_3, χ_4
[15, 40)	13.5 ± 1.5	
[40, 80)	16.0 ± 1.6	
[80, 150)	16.0 ± 2.1	

A two-parameter fit $\Delta(\ell) \approx 9.8 + 0.65\sqrt{\ell}$ has RMS 1.8 and inter-character scatter $\sim 12\%$. A dilated-Slepian reading $\lambda_k \approx (1 - \lambda_k^{\text{Slepian}})^\kappa$ produced $\kappa = 2.85 \pm 0.26$ at constrained c , then 3.46 / 4.48 / 5.37 on ten-rung ladders at free c [?]: the model has no well-defined exponent.

2 What the bands say

The four bands are not a slow square-root climb. They are a *ramp of ~ 40 nats* followed by a *plateau*:

$$\Delta(\ell) \longrightarrow \Delta_\infty \approx 16 \quad (\ell \gtrsim 40).$$

On the plateau each successive eigenvalue is larger than the previous one by a constant factor $e^{16} \approx 9 \times 10^6$. That is a geometric tail, not a Slepian tail and not a $\sqrt{\ell}$ tail.

The $\sqrt{\ell}$ fit is what a two-parameter curve does when it is asked to interpolate a step of height 5 over a run of length 40. It is a useful mnemonic. It is not the lemma.

Remark. A geometric tail $\ell_k = \ell_0 + k\Delta_\infty$ is the generic signature of a *scale-invariant leftover problem*: after the infrared desert and the turning-point region have been spent on the first rungs, each new orthogonality constraint multiplies the Rayleigh quotient by a constant. The constant is an invariant of the window, not of the zero set.

3 The lemma candidate

Definition 1 (Windowed restriction). Let V_L be the space of even real functions supported in I_L , with the L^2 inner product. For a discrete even set $\Gamma \subset \mathbb{R}$ write $Q_\Gamma(f) = \sum_{\gamma \in \Gamma} |\hat{f}(\gamma)|^2$. When Γ is the nontrivial zeros of $L(s, \chi)$ (and the pole term, for ζ), Q_Γ on V_L is the truncated Weil form, by the exact pairing of [?].

Conjecture 1 (Universality of spacings). There exists a function $\Delta_\infty : [0, \infty) \rightarrow (0, \infty)$, depending on the window construction $(V_L, \hat{\cdot})$ and not on χ , such that for every real primitive χ and every depth-adequate cutoff μ ,

$$\Delta_k(\chi, \mu) = \Delta_\infty(\ell_k(\chi, \mu)) + \varepsilon_{k, \chi, \mu},$$

with $\text{rms}(\varepsilon)/\Delta \lesssim 12\%$ on the thirty-three rungs already measured, and $\Delta_\infty(\ell) \rightarrow \Delta_\infty \approx 16$ as $\ell \rightarrow \infty$.

This is the surviving content of [?] §4, stripped of the dissolved κ -model. It is a statement in harmonic analysis: approximation numbers of a Fourier restriction to a discrete set, *after the first eigenvalue has been scaled out*. The arithmetic of χ is allowed to set λ_0 . It is not allowed to set the ratios λ_k/λ_{k+1} once depth is given.

Lemma 1 (Factorization — formal consequence). Assume Conjecture ?? and the affine ground-state law $\ell_0 = s(\chi)\mu + b(\chi)$. Then the whole quasi-null ladder is determined by two arithmetic scalars (s, b) and one window-invariant profile Δ_∞ :

$$\ell_0 = s\mu + b, \quad \ell_{k+1} = \ell_k + \Delta_\infty(\ell_k).$$

Equivalently: χ chooses a starting depth; the window chooses the staircase.

Lemma ?? is not an additional claim. It is the factorization already measured: architecture $\times$ speed. The content that still needs a proof is Conjecture ??.

4 What a proof must supply

Two harmonic-analysis assertions, neither mentioning RH.

A1. Scale invariance past the turning point. Let R_k be the Rayleigh problem for Q_Γ on the orthogonal of the first k eigenfunctions in V_L . For $\ell_k \gtrsim 40$, R_k is equivalent, up to a scalar, to R_{k+1} after a universal renormalization of V_L . In particular the minimax value drops by a factor $e^{-\Delta_\infty}$ independent of k and of Γ .

Heuristic: the first rungs spend the infrared desert (a Slepian / Airy / Landau–Widom turning point, character-dependent through γ_1). What remains is a problem on the *window* — time-limited functions, Fourier mass expelled to the band frontier (hyper-nullity [?] §16.3) — whose next mode costs a constant factor. The plateau $\Delta_\infty \approx 16$ is that factor in nats.

A2. Character-blindness of the leftover. Once λ_0 is scaled out, the leftover operator on $v_0^\perp \subset V_L$ depends on Γ only through quantities already absorbed in λ_0 . Distinct zero sets with the same window produce the same ratios λ_k/λ_{k+1} at equal depth.

Heuristic: hyper-nullity says the residual mass of a deep rung does not live on the interior zeros. If the leak is a frontier phenomenon, the interior arithmetic of Γ cannot set the next ratio.

A1 + A2 $\Rightarrow$ Conjecture ?? . Recruitment (rung k seats the $(k+1)$ -st χ -supported prime) is a statement about *organization*, not about *spacing*, and is not needed for the lemma. Mixing the two was the κ -trap.

5 What the lemma is not

- *Not* $\lambda_k = (1 - \lambda_k^{\text{Slepian}})^\kappa$. Classical prolate spacings in $-\ln(1 - \lambda)$ units are 3–5; Weil spacings are 11–19. A free κ fits any one ladder and does not transfer [?] §14.8.
- *Not* Landau–Widom for $\lambda_{\min}(L)$ as the window grows [?]. That law lives on the other axis (depth of the ground state versus L). The $s\mu$ law already occupies that axis and is linear in $\mu = e^L$, i.e. exponential in the window width — the lattice signature. $\Delta(\ell)$ is the internal mesh of one matrix.
- *Not* a function of the prime gaps. Log-gaps of primes shrink; Δ saturates. Recruitment assigns primes to rungs; spacings are not those gaps.
- *Not* a counting functional of the zeros. Vacancy is quasi-universal [?]; depth is a bandwidth phenomenon. The lemma lives on that side.

6 A numerical lock for Δ_∞

The plateau rests on two bands and a handful of deep rungs (ζ, χ_3, χ_4). Three cheap checks would freeze or kill it.

1. Ten-rung ladders at one more deep character already computed (χ_5 at $\mu = 38$): if the last four spacings sit at 16 ± 2 , the plateau is a family fact, not a ζ – χ_3 coincidence.
2. A synthetic Q_Γ with Γ a rigid unfolding of spacing 2π (no arithmetic, no desert variation): the same Δ_∞ would confirm A2 directly.
3. Dependence on L at fixed χ . If Δ_∞ is a window invariant it must be stable when μ changes and only $s\mu$ slides the starting depth. A counter-variation with L would mean Δ_∞ still carries a piece of $s(\chi)$.

Until (1)–(3) are run, $\Delta_\infty \approx 16$ is a reading of four bins, not a constant.

7 Status

Conjecture ?? is the lemma candidate. Lemma ?? is bookkeeping. A1 and A2 are the proof obligations. The $\sqrt{\ell}$ fit and the κ -model are retired as statements; they remain useful as the history of a failed identification.

Nothing here is claimed about RH. The lemma, if true, says that once the ground state has been drilled to depth $s\mu$, the rest of the quasi-null ladder is a property of the window.

References

- [1] D. Joubert, *Depth phenomenology of truncated Weil quadratic forms: fifteen L-functions, one ladder*, same repository (2026).
- [2] D. Joubert, *Proper sub-Euler products violate Weil positivity on a fixed window: certified witnesses*, same repository (2026).
- [3] D. Joubert, *Le milieu des nombres premiers*, lab notebook, same repository (2026), §14.2–14.8 and §16.3.

- [4] M. Chuk, *Weil positivity in compact windows: certified two-sided bounds and a Landau–Widom decay law*, arXiv:2608.24827 (2026).

Addendum — leakage profile, 1 September afternoon

Notebook §16.4–16.7. Rungs 0–2 share one leakage shape, scaled by depth: evidence for A1 on the zero side. D2 (a μ -independent τ) died at the 85-digit zeros; A1 is scale invariance of the *deflated* problem, not constancy of τ . The leftover still carries arithmetic through $s(\chi)$ via the endpoint law $|\hat{v}_0(\gamma_1)| \approx C\lambda_0$. A2 remains open.

Addendum — cheap locks, 1 September

Two of the three proposed locks, run.

A character already in hand. χ_{-8} at $\mu = 22$ has rungs $\ell = (25.12, 11.50)$, spacing 13.6 at level 11.5 — inside the $[0, 15)$ band 10.9 ± 1.5 of the thirty-three-rung collapse. One more L -function on the same profile, not a new architecture.

Synthetic rigid Γ . `code/delta_synthetic.py`: Q_Γ on a 2π -unfolding (no desert, no χ) at $\mu = 11$ produces spacings 7–9 nats, not 11–16. Universality across the fifteen L -functions is *not* blindness to the spacing law of Γ . A2 as written (“any discrete set, same Δ ”) fails this test. The lemma stays: one profile for the Weil forms of the family, not for every restriction to a discrete set.

The L -variation at fixed χ is the lock that still needs a depth-adequate pair of windows on a slow character (now available: χ_{-20} at $\mu = 22$ and $\mu = 32$).